

Class 12 Physics - Chapter: Electric Charges and Fields

Practice Questions with Solutions

Q1: Which of the following is NOT a property of electric field lines?

- a) They originate from positive charges and terminate on negative charges.
- b) They never intersect each other.
- c) They form closed loops.
- d) They are denser where the electric field is stronger.

Answer: c)

Solution: Electric field lines never form closed loops. They start from positive charges and end on negative charges or extend to infinity.



Q2: What is the SI unit of electric charge?

- a) Coulomb
- b) Ampere
- c) Volt
- d) Ohm

Answer: a)

Solution: The SI unit of electric charge is Coulomb (C).

Q3: The force between two charges placed in vacuum is inversely proportional to:

- a) Product of charges
- b) Square of distance between them

- c) Sum of charges
- d) Square root of distance between them

Answer: b)

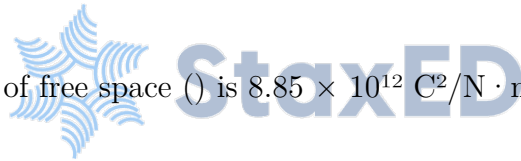
Solution: According to Coulomb's law, force is inversely proportional to the square of the distance between the charges.

Q4: The value of the permittivity of free space () is approximately:

- a) $8.85 \times 10^{12} \text{ C}^2/\text{N} \cdot \text{m}^2$
- b) $9 \times 10 \text{ Nm}^2/\text{C}^2$
- c) $1.6 \times 10^1 \text{ C}$
- d) $3 \times 10 \text{ m/s}$

Answer: a)

Solution: The permittivity of free space () is $8.85 \times 10^{12} \text{ C}^2/\text{N} \cdot \text{m}^2$.



Q5: Which instrument is used to detect electric charge?

- a) Ammeter
- b) Voltmeter
- c) Electroscope
- d) Galvanometer

Answer: c)

Solution: An electroscope is used to detect the presence of electric charge on a body.

Q6: The electric field due to a point charge is:

- a) Uniform
- b) Constant in magnitude

- c) Radially outward or inward
- d) Tangential to the direction of motion

Answer: c)

Solution: The electric field lines due to a point charge are radial — they point outward for a positive charge and inward for a negative charge.

Q7: Two point charges $+q$ and $-q$ placed at a distance form:

- a) A monopole
- b) A dipole
- c) A quadrupole
- d) A neutral system

Answer: b)

Solution: A pair of equal and opposite charges separated by a small distance forms an electric dipole.



Q8: Which quantity is a scalar?

- a) Electric force
- b) Electric field
- c) Electric flux
- d) Electric potential gradient

Answer: c)

Solution: Electric flux is a scalar quantity while electric field and force are vectors.

Q9: The superposition principle is applicable to:

- a) Scalar quantities only

- b) Vector quantities only
- c) Both scalar and vector quantities
- d) Neither scalar nor vector quantities

Answer: b)

Solution: The superposition principle applies to vector quantities like electric field and force.

Q10: Which of the following affects the force between two charges?

- a) Medium between charges
- b) Distance between charges
- c) Magnitude of charges
- d) All of the above

Answer: d)



Solution: All three — medium, distance, and magnitude — influence the electrostatic force.

Q11: State the principle of superposition of electric fields.

Solution: The net electric field due to a group of point charges is the vector sum of the fields due to individual charges.

Q12: What is the significance of the direction of electric field lines?

Solution: The direction of electric field lines indicates the direction of the force experienced by a positive test charge placed in the field.

Q13: Can electric field exist without charges? Explain.

Solution: No, electric fields are produced by electric charges. However, fields can exist in space due to nearby charges even if no charge is present at that point.

Q14: What is the nature of electric field at the midpoint of a dipole?

Solution: The electric field at the midpoint of a dipole is directed opposite to the direction of the dipole moment and has a magnitude proportional to $1/r^3$.

Q15: What is the SI unit of electric field?

Solution: The SI unit of electric field is Newton per Coulomb (N/C).

Q16: Give one application of Gauss's Law.

Solution: Gauss's Law is used to calculate electric field of a uniformly charged spherical shell or sphere.

Q17: Define an electric dipole.

Solution: An electric dipole is a pair of equal and opposite charges separated by a small fixed distance.

Q18: What is the relation between electric field and electric potential?

Solution: Electric field is the negative gradient of electric potential, i.e., $E = -dV/dx$.

Q19: Can two electric field lines cross each other? Why or why not?

Solution: No, electric field lines never cross because at any point in space, the electric field must have a unique direction.

Q20: State Coulomb's law and define 1 Coulomb of charge.

Solution: Coulomb's Law: The force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them. 1 Coulomb is the amount of charge that exerts a force of 9×10^9 N on an equal charge placed 1 meter apart in vacuum.

Q21: What is an electric dipole? Derive expressions for the electric field at a point (a) on the axial line and (b) on the equatorial line of a dipole.

Solution: An electric dipole consists of two equal and opposite charges separated by a small distance. Dipole moment $p = q \times 2a$. (a) Axial Line: $E = \frac{1}{4\pi\epsilon_0} \times \frac{2p}{r^3}$ (b) Equatorial Line: $E = \frac{1}{4\pi\epsilon_0} \times \frac{p}{r^3}$, opposite to dipole moment. The electric field decreases with the cube of the distance from the dipole.

Q22: Explain the concept of electric flux. Derive its expression and explain its significance with examples.

Solution: Electric flux is the total number of electric field lines passing normally through a surface. $\Phi = E \cdot A \cdot \cos\theta$. Significance: Helps quantify electric field interactions with surfaces and is key in Gauss's law. Example: Flux through a sphere enclosing a point charge q is q/ϵ_0 .

Q23: State and prove Gauss's Law. List its applications.

Solution: Gauss's Law: Net electric flux through a closed surface equals $1/\epsilon_0$ times the charge enclosed. $\oint \mathbf{E} \cdot d\mathbf{A} = Q_{\text{enclosed}} / \epsilon_0$. Proof: Using a point charge and symmetry. Applications:
1. Field due to infinite line charge 2. Field due to plane sheet 3. Field due to spherical shell

Q24: Describe the electric field due to a uniformly charged infinite plane sheet. Derive its expression using Gauss's law.

Solution: Use a Gaussian 'pillbox' that cuts through the plane. Field emerges perpendicular from both surfaces. Flux = $2EA$, charge enclosed = QA Gauss's Law: $2EA = QA/\epsilon_0 \rightarrow E = Q/(2\epsilon_0)$, direction is normal to the plane.

Q25: Explain how electric field behaves inside and outside a uniformly charged spherical shell using Gauss's law.

Solution: Outside ($r > R$): $E = (1/4\pi\epsilon_0) \cdot (Q/r^2)$, radial. Inside ($r < R$): $E = 0$ as enclosed charge is zero. Gauss's surface does not include any charge inside the shell, hence no field.

Q26: What are the differences between electric field and electric potential?

Solution: 1. Electric field is a vector; potential is scalar. 2. $E = -dV/dx$ 3. Field shows force per unit charge; potential shows energy per unit charge. 4. Units: $E \rightarrow \text{N/C}$, $V \rightarrow \text{Volt}$. 5. Field lines give direction; equipotential lines are perpendicular to field lines.

Q27: Describe the concept of electric shielding and its practical applications.

Solution: Electric shielding prevents external electric fields from affecting a region. Achieved using a conductor. Inside a conductor, $E = 0$. Applications: 1. Shielding sensitive electronic devices 2. Faraday cages 3. Coaxial cables

Q28: What is the role of permittivity in electrostatics? Explain relative permittivity.

Solution: Permittivity (ϵ) determines how much electric field is allowed in a medium. Force in medium: $F = (1 / 4\pi\epsilon) \cdot (qq / r^2)$. Relative permittivity $\epsilon_r = \epsilon / \epsilon_0$. High ϵ_r reduces electrostatic force.

Q29: Explain with diagram how electric field lines behave in the vicinity of two like and unlike point charges.

Solution: Like charges: Field lines diverge, show repulsion. Unlike charges: Lines start at $+q$ and end at $-q$, showing attraction. At midpoints, directions cancel in like charges and add in unlike charges.

Q30: Calculate the electric field at a point 0.5 m from a 2 C point charge in air.

Solution: $E = (1 / 4\pi\epsilon_0) \cdot (q / r^2) = 9 \times 10^9 \times (2 \times 10) / (0.5)^2 = 72,000 \text{ N/C}$

Q31: Find the force between two charges +5 C and +10 C placed 10 cm apart in vacuum.

Solution: $F = (1 / 4\pi\epsilon_0) \cdot (qq / r^2) = 9 \times 10^9 \times (5 \times 10)(10 \times 10) / (0.1)^2 = 45 \text{ N}$

Q32: What is the electric flux through a spherical surface enclosing a 2 mC charge?

Solution: $\Phi = q / \epsilon_0 = (2 \times 10^{-3}) / (8.85 \times 10^{-12}) = 2.26 \times 10^8 \text{ Nm}^2/\text{C}$

Q33: A 6 C charge is placed in a uniform electric field of strength $4 \times 10^4 \text{ N/C}$. Find the force acting on it.

Solution: $F = qE = 6 \times 10 \times 4 \times 10^4 = 0.24 \text{ N}$

Q34: A charge of 3 C moves through a potential difference of 12 V. Calculate the work done.

Solution: $W = qV = 3 \times 10 \times 12 = 3.6 \times 10^2 \text{ J}$

Q35: Determine the electric field intensity due to a line charge of 5 C/m at a distance of 2 m.

Solution: $E = (1 / 2\pi\epsilon_0) \cdot (q / r) = (9 \times 10^9 / 2) \times (5 \times 10 / 2) = 3.58 \times 10^5 \text{ N/C}$

Q36: A square of side 10 cm has a charge of 4 C uniformly distributed. Find electric flux through one face.

Solution: Total flux $= q / \epsilon_0 = 4 \times 10 / 8.85 \times 10^{-12} = 4.52 \times 10^8 \text{ Nm}^2/\text{C}$. One face gets 1/6th: $7.54 \times 10^7 \text{ Nm}^2/\text{C}$

Q37: Calculate the electric field inside a conductor carrying a static charge.

Solution: Inside a conductor with static charge, the electric field is always zero.

Q38: A proton experiences a force of 3.2×10^1 N in a field. Find the field strength.

Solution: $q = 1.6 \times 10^1$ C; $E = F / q = 3.2 \times 10^1 / 1.6 \times 10^1 = 2 \times 10$ N/C

Q39: A cube encloses a point charge of 6 C. Find total electric flux through the cube.

Solution: $\Phi = q / \epsilon_0 = 6 \times 10 / 8.85 \times 10^{12} = 6.78 \times 10$ Nm²/C

